

Cryptography, Cryptanalysis, and Out of Scope Definitions:

(1) **Cryptography paper** - a paper that is *centered* around constructing or designing a novel cryptographic object or cryptosystem, with analysis of extant cryptosystems as related or prior work. Look for: 1) a definition of security, 2) a proof that a construction meets the given security definition.

Note: Papers with formal analysis as proofs should be marked as confused and brought to the entire team.

(2) **Cryptanalysis paper** - a paper that is *centered* around analyzing an existing cryptographic object or cryptosystem—possibly identifying flaws—with ideas or designs for improving cryptographic objects or cryptosystems as potential secondary contributions. Look for: 1) identification of a particular cryptographic object or cryptosystem and 2) a method of analysis—which can be theoretical (referring to mathematical definitions and proofs) and/or practical (referring to software and hardware implementation)—that demonstrates potential or lived vulnerabilities in that cryptographic object or cryptosystem.

Note: Papers that give new proofs for already deployed protocols are cryptanalysis papers.

(3) **Out of scope papers** - Papers that do provide some sort of security analysis but said analysis is not a proof that is not a formal mathematical analysis of the security model. Not cryptography/cryptanalysis paper.